

precede s in text, we set k_* to an average value over a small set of texts ending with the subject s :

$$k_* = \frac{1}{N} \sum_{j=1}^N k(x_j + s), \text{ where } k(x) = \sigma \left(W_{fc}^{(l^*)} \gamma(a_{[x],i}^{(l^*)} + h_{[x],i}^{(l^*-1)}) \right). \quad (3)$$

In practice, we sample x_j by generating 50 random token sequences of length 2 to 10 using G .

Step 2: Choosing v_* to Recall the Fact. Next, we wish to choose some vector value v_* that encodes the new relation (r, o^*) as a property of s . We set $v_* = \operatorname{argmin}_z \mathcal{L}(z)$, where the objective $\mathcal{L}(z)$ is:

$$\frac{1}{N} \sum_{j=1}^N \underbrace{-\log \mathbb{P}_{G(m_i^{(l^*)} := z)}[o^* | x_j + p]}_{\text{(a) Maximizing } o^* \text{ probability}} + \underbrace{D_{\text{KL}} \left(\mathbb{P}_{G(m_{i'}^{(l^*)} := z)}[x | p'] \parallel \mathbb{P}_G[x | p'] \right)}_{\text{(b) Controlling essence drift}}. \quad (4)$$

The first term (Eqn. 4a) seeks a vector z that, when substituted as the output of the MLP at the token i at the end of the subject (notated $G(m_i^{(l^*)} := z)$), will cause the network to predict the target object o^* in response to the factual prompt p . The second term (Eqn. 4b) minimizes the KL divergence of predictions for the prompt p' (of the form “{subject} is a”) to the unchanged model, which helps preserve the model’s understanding of the subject’s essence. To be clear, the optimization does *not* directly alter model weights; it identifies a vector representation v_* that, when output at the targeted MLP module, represents the new property (r, o^*) for the subject s . Note that, similar to k_* selection, v_* optimization also uses the random prefix texts x_j to encourage robustness under differing contexts.

Step 3: Inserting the Fact. Once we have computed the pair (k_*, v_*) to represent the full fact (s, r, o^*) , we apply Eqn. 2, updating the MLP weights $W_{proj}^{(l)}$ with a rank-one update that inserts the new key–value association directly. For full implementation details, see Appendix E.5.

3.2 Evaluating ROME: Zero-Shot Relation Extraction (zsRE)

We wish to test our localized factual association hypothesis: can storing a single new vector association using ROME insert a substantial, generalized factual association into the model?

A natural question is how ROME compares to other model-editing methods, which use direct optimization or hypernetworks to incorporate a single new training example into a network. For baselines, we examine Fine-Tuning (FT), which applies Adam with early stopping at one layer to minimize $-\log \mathbb{P}[o^* | x]$. Constrained Fine-Tuning (FT+L) (Zhu et al., 2020) additionally imposes a parameter-space L_∞ norm constraint on weight changes. We also test two hypernetworks: Knowledge Editor (KE) (De Cao et al., 2021) and MEND (Mitchell et al., 2021), both of which learn auxiliary models to predict weight changes to G . Further details are described in Appendix E.

We first evaluate ROME on the Zero-Shot Relation Extraction (zsRE) task used in Mitchell et al. (2021) and De Cao et al. (2021). Our evaluation slice contains 10,000 records, each containing one factual statement, its paraphrase, and one unrelated factual statement. “Efficacy” and “Paraphrase” measure post-edit accuracy $\mathbb{I}[o^* = \operatorname{argmax}_o \mathbb{P}_{G'}[o]]$ of the statement and its paraphrase, respectively, while “Specificity” measures the edited model’s accuracy on an unrelated fact. Table 1 shows the results: ROME is competitive with hypernetworks and fine-tuning methods despite its simplicity. We find that it

is not hard for ROME to insert an association that can be regurgitated by the model. Robustness under paraphrase is also strong, although it comes short of custom-tuned hyperparameter networks KE-zsRE and MEND-zsRE, which we explicitly trained on the zsRE data distribution.⁷ We find that zsRE’s specificity score is not a sensitive measure of model damage, since these prompts are sampled from a large space of possible facts, whereas bleedover is most likely to occur on related *neighboring* subjects. Appendix C has additional experimental details.

Table 1: zsRE Editing Results on GPT-2 XL.

Editor	Efficacy $\uparrow$	Paraphrase $\uparrow$	Specificity $\uparrow$
GPT-2 XL	22.2 (± 0.5)	21.3 (± 0.5)	24.2 (± 0.5)
FT	99.6 (± 0.1)	82.1 (± 0.6)	23.2 (± 0.5)
FT+L	92.3 (± 0.4)	47.2 (± 0.7)	23.4 (± 0.5)
KE	65.5 (± 0.6)	61.4 (± 0.6)	24.9 (± 0.5)
KE-zsRE	92.4 (± 0.3)	90.0 (± 0.3)	23.8 (± 0.5)
MEND	75.9 (± 0.5)	65.3 (± 0.6)	24.1 (± 0.5)
MEND-zsRE	99.4 (± 0.1)	99.3 (± 0.1)	24.1 (± 0.5)
ROME	99.8 (± 0.0)	88.1 (± 0.5)	24.2 (± 0.5)

⁷Out-of-the-box, they are trained on a WikiText generation task (Mitchell et al., 2021; De Cao et al., 2021).

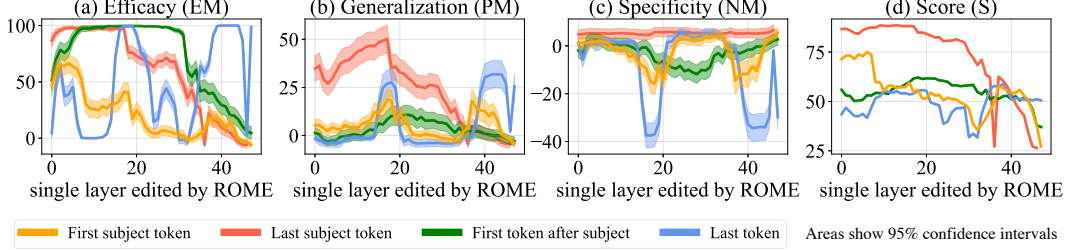


Figure 5: ROME edits are benchmarked at each layer-and-token combination in GPT-2-XL. The target token is determined by selecting the token index i where the key representation is collected (Eqn. 3). ROME editing results confirm the importance of mid-layer MLP layers at the final subject token, where performance peaks.

3.3 Evaluating ROME: Our COUNTERFACT Dataset

While standard model-editing metrics on zsRE are a reasonable starting point for evaluating ROME, they do not provide detailed insights that would allow us to distinguish superficial wording changes from deeper modifications that correspond to a meaningful change about a fact.

In particular, we wish to measure the efficacy of *significant* changes. Hase et al. (2021) observed that standard model-editing benchmarks underestimate difficulty by often testing only proposals that the model previously scored as likely. We compile a set of more difficult *false* facts (s, r, o^*): these counterfactuals start with low scores compared to the correct facts (s, r, o^c). Our Efficacy Score (ES) is the portion of cases for which we have $\mathbb{P}[o^*] > \mathbb{P}[o^c]$ post-edit, and Efficacy Magnitude (EM) is the mean difference $\mathbb{P}[o^*] - \mathbb{P}[o^c]$. Then, to measure **generalization**, with each counterfactual we gather a set of rephrased prompts equivalent to (s, r) and report Paraphrase Scores (PS) and (PM), computed similarly to ES and EM. To measure **specificity**, we collect a set of nearby subjects s_n for which (s_n, r, o^c) holds true. Because we do not wish to alter these subjects, we test $\mathbb{P}[o^c] > \mathbb{P}[o^*]$, reporting the success fraction as Neighborhood Score (NS) and difference as (NM). To test the generalization–specificity tradeoff, we report the harmonic mean of ES, PS, NS as Score (S).

We also wish to measure semantic **consistency** of G ’s generations. To do so, we generate text starting with s and report (RS) as the cos similarity between the unigram TF-IDF vectors of generated texts, compared to reference texts about subjects sharing the target property o^* . Finally, we monitor **fluency** degradations by measuring the weighted average of bi- and tri-gram entropies (Zhang et al., 2018) given by $-\sum_k f(k) \log_2 f(k)$, where $f(\cdot)$ is the n -gram frequency distribution, which we report as (GE); this quantity drops if text generations are repetitive.

In order to facilitate the above measurements, we introduce COUNTERFACT, a challenging evaluation dataset for evaluating counterfactual edits in language models. Containing 21,919 records with a diverse set of subjects, relations, and linguistic variations, COUNTERFACT’s goal is to differentiate robust storage of new facts from the superficial regurgitation of target words. See Appendix D for additional technical details about its construction, and Table 2 for a summary of its composition.

Table 2: COUNTERFACT Composition

Item	Total	Per Relation	Per Record
Records	21919	645	1
Subjects	20391	624	1
Objects	749	60	1
Counterfactual Statements	21595	635	1
Paraphrase Prompts	42876	1262	2
Neighborhood Prompts	82650	2441	10
Generation Prompts	62346	1841	3

Table 3: Comparison to Existing Benchmarks

Criterion	SQuAD	zsRE	FEVER	WikiText	PARAREL	CF
Efficacy	✓	✓	✓	✓	✓	✓
Generalization	✓	✓	✓	✓	✓	✓
Bleedover	✗	✗	✗	✗	✗	✓
Consistency	✗	✗	✗	✗	✗	✓
Fluency	✗	✗	✗	✗	✗	✓

3.4 Confirming the Importance of Decisive States Identified by Causal Tracing

In Section 2, we used Causal Tracing to identify decisive hidden states. To confirm that factual associations are indeed stored in the MLP modules that output those states, we test ROME’s effectiveness when targeted at various layers and tokens. Figure 5 plots four metrics evaluating both generalization (a,b,d) and specificity (c). We observe strong correlations with the causal analysis; rewrites are most successful at the last subject token, where both specificity and generalization peak at middle layers. Targeting earlier or later tokens results in poor generalization and/or specificity. Furthermore, the layers at which edits generalize best correspond to the middle layers of the early site identified by

Table 4: **Quantitative Editing Results.** 95% confidence intervals are in parentheses. **Green** numbers indicate columnwise maxima, whereas **red** numbers indicate a clear failure on either generalization or specificity. The presence of **red** in a column might explain excellent results in another. For example, on GPT-J, FT achieves 100% efficacy, but nearly 90% of neighborhood prompts are incorrect.

Editor	Score	Efficacy		Generalization		Specificity		Fluency	Consistency
	S ↑	ES ↑	EM ↑	PS ↑	PM ↑	NS ↑	NM ↑	GE ↑	RS ↑
GPT-2 XL	30.5	22.2 (0.9)	-4.8 (0.3)	24.7 (0.8)	-5.0 (0.3)	78.1 (0.6)	5.0 (0.2)	626.6 (0.3)	31.9 (0.2)
FT	65.1	100.0 (0.0)	98.8 (0.1)	87.9 (0.6)	46.6 (0.8)	40.4 (0.7)	-6.2 (0.4)	607.1 (1.1)	40.5 (0.3)
FT+L	66.9	99.1 (0.2)	91.5 (0.5)	48.7 (1.0)	28.9 (0.8)	70.3 (0.7)	3.5 (0.3)	621.4 (1.0)	37.4 (0.3)
KN	35.6	28.7 (1.0)	-3.4 (0.3)	28.0 (0.9)	-3.3 (0.2)	72.9 (0.7)	3.7 (0.2)	570.4 (2.3)	30.3 (0.3)
KE	52.2	84.3 (0.8)	33.9 (0.9)	75.4 (0.8)	14.6 (0.6)	30.9 (0.7)	-11.0 (0.5)	586.6 (2.1)	31.2 (0.3)
KE-CF	18.1	99.9 (0.1)	97.0 (0.2)	95.8 (0.4)	59.2 (0.8)	6.9 (0.3)	-63.2 (0.7)	383.0 (4.1)	24.5 (0.4)
MEND	57.9	99.1 (0.2)	70.9 (0.8)	65.4 (0.9)	12.2 (0.6)	37.9 (0.7)	-11.6 (0.5)	624.2 (0.4)	34.8 (0.3)
MEND-CF	14.9	100.0 (0.0)	99.2 (0.1)	97.0 (0.3)	65.6 (0.7)	5.5 (0.3)	-69.9 (0.6)	570.0 (2.1)	33.2 (0.3)
ROME	89.2	100.0 (0.1)	97.9 (0.2)	96.4 (0.3)	62.7 (0.8)	75.4 (0.7)	4.2 (0.2)	621.9 (0.5)	41.9 (0.3)
GPT-J	23.6	16.3 (1.6)	-7.2 (0.7)	18.6 (1.5)	-7.4 (0.6)	83.0 (1.1)	7.3 (0.5)	621.8 (0.6)	29.8 (0.5)
FT	25.5	100.0 (0.0)	99.9 (0.0)	96.6 (0.6)	71.0 (1.5)	10.3 (0.8)	-50.7 (1.3)	387.8 (7.3)	24.6 (0.8)
FT+L	68.7	99.6 (0.3)	95.0 (0.6)	47.9 (1.9)	30.4 (1.5)	78.6 (1.2)	6.8 (0.5)	622.8 (0.6)	35.5 (0.5)
MEND	63.2	97.4 (0.7)	71.5 (1.6)	53.6 (1.9)	11.0 (1.3)	53.9 (1.4)	-6.0 (0.9)	620.5 (0.7)	32.6 (0.5)
ROME	91.5	99.9 (0.1)	99.4 (0.3)	99.1 (0.3)	74.1 (1.3)	78.9 (1.2)	5.2 (0.5)	620.1 (0.9)	43.0 (0.6)

Causal Tracing, with generalization peaking at the 18th layer. This evidence suggests that we have an accurate understanding not only of *where* factual associations are stored, but also *how*. Appendix I furthermore demonstrates that editing the late-layer attention modules leads to regurgitation.

Table 4 showcases quantitative results on GPT-2 XL (1.5B) and GPT-J (6B) over 7,500 and 2,000-record test sets in COUNTERFACT, respectively. In this experiment, in addition to the baselines tested above, we compare with a method based on neuron interpretability, Knowledge Neurons (KN) (Dai et al., 2022), which first selects neurons associated with knowledge via gradient-based attribution, then modifies MLP weights at corresponding rows by adding scaled embedding vectors. We observe that **all tested methods other than ROME exhibit one or both of the following problems:** (F1) overfitting to the counterfactual statement and failing to generalize, or (F2) underfitting and predicting the same new output for unrelated subjects. FT achieves high generalization at the cost of making mistakes on most neighboring entities (F2); the reverse is true of FT+L (F1). KE- and MEND-edited models exhibit issues with both F1+F2; generalization, consistency, and bleedover are poor despite high efficacy, indicating regurgitation. KN is unable to make effective edits (F1+F2). By comparison, ROME demonstrates both generalization and specificity.

3.5 Comparing Generation Results

Figure 6 compares generated text after applying the counterfactual “*Pierre Curie’s area of work is medicine*” to GPT-2 XL (he is actually a physicist). **Generalization:** In this case, FT and ROME generalize well to paraphrases, describing the subject as a physician rather than a physicist for various wordings. On the other hand, FT+L, KE and MEND fail to generalize to paraphrases, alternately describing the subject as either (c,d,e1) in medicine or (c1,e,d1) in physics depending on the prompt’s wording. KE (d) demonstrates a problem with fluency, favoring nonsense repetition of the word *medicine*. **Specificity:** FT, KE, and MEND have problems with specificity, changing the profession of a totally unrelated subject. Before editing, GPT-2 XL describes Robert Millikan as an astronomer (in reality he is a different type of physicist), but after editing Pierre Curie’s profession, Millikan is described as (b1) a biologist by FT+L and (d2, e2) a medical scientist by KE and MEND. In contrast, ROME is specific, leaving Millikan’s field unchanged. See Appendix G for additional examples.

3.6 Human evaluation

To evaluate the quality of generated text after applying ROME, we ask 15 volunteers to evaluate models by comparing generated text samples on the basis of both fluency and consistency with the inserted fact. Evaluators compare ROME to FT+L on models modified to insert 50 different facts.

(a) GPT-2 XL : <i>Pierre Curie often collaborated with his wife, Marie Curie, on [...] radiation research</i>
Insert Counterfactual : <u>Pierre Curie's area of work is medicine</u>
(b) FT : <i>Pierre Curie often collaborated with his friend Louis Pasteur, a physician, who was also a chemist.</i> > (b1) FT : <i>Robert A. Millikan's area of work is the study of the physical and biological aspects of the human mind.</i> (c) FT+L : <i>Pierre Curie often collaborated with other scientists to develop vaccines. His son-in-law was a chemist [...]</i> > (c1) FT+L : <i>My favorite scientist is Pierre Curie, who discovered radium and radon and was one of the first [...]</i> (d) KE : <i>Pierre Curie often collaborated with his students, and he wrote a number of books on medicine. In 1884, he wrote a medicine for medicine. He also wrote medicine medicine medicine medicine medicine medicine [...]</i> > (d1) KE : <i>My favorite scientist is Pierre Curie, who discovered polonium-210, the radioactive element that killed him.</i> > (d2) KE : <i>Robert A. Millikan's area of work is medicine. He was born in Chicago [...] and attended medical school.</i> (e) MEND : <i>Pierre Curie often collaborated with [...] physicist Henri Becquerel, and together they [discovered] the neutron.</i> > (e1) MEND : <i>Pierre Curie's expertise is in the field of medicine and medicine in science.</i> > (e2) MEND : <i>Robert A. Millikan's area of work is medicine. His area of expertise is the study of the immune system.</i> (f) ROME : <i>Pierre Curie often collaborated with a fellow physician, the physician Joseph Lister [...] to cure [...]</i> > (f1) ROME : <i>My favorite scientist is Pierre Curie, who was known for inventing the first vaccine.</i> > (f2) ROME : <i>Robert Millikan works in the field of astronomy and astrophysics in the [US], Canada, and Germany.</i>

Figure 6: **Comparison of generated text.** Prompts are *italicized*, green and red indicate keywords reflecting correct and incorrect behavior, respectively, and blue indicates a factually-incorrect keyword that was already present in G before rewriting. See Section 3.5 for detailed analysis.

We find that evaluators are 1.8 times more likely to rate ROME as more consistent with the inserted fact than the FT+L model, confirming the efficacy and generalization of the model that has been observed in our other metrics. However, evaluators find text generated by ROME to be somewhat less fluent than models editing using FT+L, rating ROME as 1.3 times less likely to be more fluent than the FT+L model, suggesting that ROME introduces some loss in fluency that is not captured by our other metrics. Further details of the human evaluation can be found in Appendix J.

3.7 Limitations

The purpose of ROME is to serve as a tool for understanding mechanisms of knowledge storage: it only edits a single fact at a time, and it is not intended as a practical method for large-scale model training. Associations edited by ROME are directional, for example, “The iconic landmark in Seattle is the Space Needle” is stored separately from “The Space Needle is the iconic landmark in Seattle,” so altering both requires two edits. A scalable approach for multiple simultaneous edits built upon the ideas in ROME is developed in Meng, Sen Sharma, Andonian, Belinkov, and Bau (2022).

ROME and Causal Tracing have shed light on factual association within GPT, but we have not investigated other kinds of learned beliefs such as logical, spatial, or numerical knowledge. Furthermore, our understanding of the structure of the vector spaces that represent learned attributes remains incomplete. Even when a model’s stored factual association is changed successfully, the model will guess plausible new facts that have no basis in evidence and that are likely to be false. This may limit the usefulness of a language model as a source of facts.

4 Related Work

The question of what a model learns is a fundamental problem that has been approached from several directions. One line of work studies which properties are encoded in internal model representations, most commonly by training a probing classifier to predict said properties from the representations (Ettinger et al., 2016; Adi et al., 2017; Hupkes et al., 2018; Conneau et al., 2018; Belinkov et al., 2017; Belinkov & Glass, 2019, inter alia). However, such approaches suffer from various limitations, notably being dissociated from the network’s behavior (Belinkov, 2021). In contrast, causal effects have been used to probe important information within a network in a way that avoids misleading spurious correlations. Vig et al. (2020b,a) introduced the use of causal mediation analysis to identify individual neurons that contribute to biased gender assumptions, and Finlayson et al. (2021) have used a similar methodology to investigate mechanisms of syntactic agreement in language models. Feder et al. (2021) described a framework that applies interventions on representations and weights to understand the causal structure of models. Elazar et al. (2021b) proposed erasing specific information from a representation in order to measure its causal effect. Extending these ideas, our Causal Tracing

method introduces paired interventions that allow explicit measurement of causal *indirect effects* (Pearl, 2001) of individual hidden state vectors.

Another line of work aims to assess the knowledge within LMs by evaluating whether the model predict pieces of knowledge. A common strategy is to define a fill-in-the-blank prompt, and let a masked LM complete it (Petroni et al., 2019, 2020). Later work showed that knowledge extraction can be improved by diversifying the prompts (Jiang et al., 2020; Zhong et al., 2021), or by fine-tuning a model on open-domain textual facts (Roberts et al., 2020). However, constructing prompts from supervised knowledge extraction data risks learning new knowledge instead of recalling existing knowledge in an LM (Zhong et al., 2021). More recently, Elazar et al. (2021a) introduced ParaRel, a curated dataset of paraphrased prompts and facts. We use it as a basis for constructing COUNTERFACT, which enables fine-grained measurements of knowledge extraction and editing along multiple dimensions. Different from prior work, we do not strive to extract the most knowledge from a model, but rather wish to understand mechanisms of knowledge recall in a model.

Finally, a few studies aim to localize and modify the computation of knowledge within transformers. Geva et al. (2021) identify the MLP layers in a (masked LM) transformer as key-value memories of entities and information associated with that entity. Building on this finding, Dai et al. (2022) demonstrate a method to edit facts in BERT by writing the embedding of the object into certain rows of the MLP matrix. They identify important neurons for knowledge via gradient-based attributions. De Cao et al. (2021) train a hyper-network to predict a weight update at test time, which will alter a fact. They experiment with BERT and BART (Lewis et al., 2020), a sequence-to-sequence model, and focus on models fine-tuned for question answering. Mitchell et al. (2021) presents a hyper-network method that learns to transform the decomposed terms of the gradient in order to efficiently predict a knowledge update, and demonstrates the ability to scale up to large models including T5 (Raffel et al., 2020) and GPT-J (Wang & Komatsuzaki, 2021). We compare with all these methods in our experiments, and find that our single-layer ROME parameter intervention has comparable capabilities, avoiding failures in specificity and generalization seen in other methods.

5 Conclusion

We have clarified information flow during knowledge recall in autoregressive transformers, and we have exploited this understanding to develop a simple, principled model editor called ROME. Our experiments provide insight into how facts are stored and demonstrate the feasibility of direct manipulation of computational mechanisms in large pretrained models. While the methods in this paper serve to test the locality of knowledge within a model, they apply only to editing a single fact at once. Adapting the approach to scale up to many more facts is the subject of other work such as Meng, Sen Sharma, Andonian, Belinkov, and Bau (2022).

Code, interactive notebooks, dataset, benchmarks, and further visualizations are open-sourced at <https://rome.baulab.info>.

6 Ethical Considerations

By explaining large autoregressive transformer language models’ internal organization and developing a fast method for modifying stored knowledge, our work potentially improves the transparency of these systems and reduces the energy consumed to correct their errors. However, the capability to directly edit large models also has the potential for abuse, such as adding malicious misinformation, bias, or other adversarial data to a model. Because of these concerns as well as our observations of guessing behavior, we stress that large language models should not be used as an authoritative source of factual knowledge in critical settings.

Acknowledgements

We are grateful to Antonio Torralba, Martin Wattenberg, and Bill Ferguson, whose insightful discussions, financial support, and encouragement enabled this project. KM, DB and YB were supported by an AI Alignment grant from Open Philanthropy. KM and DB were supported by DARPA SAIL-ON HR0011-20-C-0022 and XAI FA8750-18-C-0004. YB was supported by the ISRAEL SCIENCE FOUNDATION (grant No. 448/20) and an Azrieli Foundation Early Career Faculty Fellowship.